LIDOCAINE

Check max dose by weight

High-alert IV

CHEMICAL NAME 2-(diethylamino)-N-(2,6-dimethylphenyl)acetamide (as hydrochloride)
CLASS Amide local anaesthetic; class Ib antiarrhythmic
TRADE NAMES Xylocaine · Lignocaine · EMLA (with prilocaine) · Lidoderm (patch)

THERAPEUTIC USES / INDICATIONS

- Local infiltration, nerve blocks, epidural and spinal anaesthesia.
- Topical anaesthesia: gel for catheterisation, spray for airway, patches for neuropathic pain.
- Ventricular arrhythmias (VT/VF); second-line to amiodarone.
- Reducing pain of propofol injection or IV cannulation.

ADMINISTRATION

- **Infiltration:** aspirate before injecting to avoid intravascular injection.
- **With adrenaline:** prolongs action; never in fingers, toes, penis, nose, ears (end-arteries).
- **IV antiarrhythmic:** bolus then infusion with cardiac monitoring.
- **Topical:** gel 2% for urethral catheterisation — wait 3–5 min.

DOSAGES

- **Local plain:** max 3 mg/kg (max 200 mg).
- **With adrenaline:** max 7 mg/kg (max 500 mg).
- **Arrhythmia:** 1–1.5 mg/kg IV bolus; infusion 1–4 mg/min.
- 1% = 10 mg/mL; 2% = 20 mg/mL — calculate carefully.

PHARMACOKINETICS

- **Onset:** infiltration 1–2 min; IV immediate. **Duration:** 1–2 h (2–4 h with adrenaline).
- Metabolised rapidly in liver (CYP1A2, 3A4).
- Excreted in urine. **t½:** 1.5–2 h (longer in heart/liver failure).

PHARMACODYNAMICS

- Blocks **voltage-gated sodium channels** in nerve membranes → no action potential → reversible loss of sensation.
- In heart: shortens action potential in ventricular tissue, suppresses ectopic activity.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Local burning on injection.
- Numbness beyond the area; transient hypotension.
- Light-headedness, tinnitus, perioral tingling (early toxicity).

ADVERSE EFFECTS (SERIOUS)

- Local anaesthetic systemic toxicity (LAST): metallic taste, tinnitus, confusion → seizures → coma, arrhythmias, cardiac arrest. Treatment: stop injection, airway, 20% lipid emulsion.
- Bradycardia, heart block.
- Methaemoglobinaemia (topical, infants).
- Allergy (rare with amides).

CONTRAINDICATIONS & CAUTIONS

- Allergy to amide local anaesthetics.
- Complete heart block, severe SA or AV block (IV use).
- Adrenaline-containing forms in end-artery areas.
- **Caution:** heart failure, liver disease, elderly, epilepsy.

KEY INTERACTIONS

- **β-blockers, cimetidine:** ↓ lidocaine clearance.
- **Other antiarrhythmics:** additive cardiac effects.

NURSING MANAGEMENT

- Confirm concentration and calculate max dose by weight.
- Ask about allergy; check for adrenaline in solution.
- Monitor ECG, BP and neurological status with IV use.
- Watch for early toxicity signs: tingling lips, tinnitus, confusion.
- Protect numb area from injury; nil by mouth until throat sensation returns after airway spray.

PATIENT EDUCATION

- Numbness lasts 1–3 hours; avoid hot drinks and chewing.
- Report dizziness, ringing in ears or metallic taste.
- Do not apply patches to broken skin; remove after 12 h.

KEY POINTS

- Na^+ channel blocker — local anaesthetic and class Ib antiarrhythmic.
- Toxicity: perioral tingling → seizures → arrest; lipid emulsion.
- Max 3 mg/kg plain; 7 mg/kg with adrenaline.
- No adrenaline in digits, nose, ears, penis.